

TREXQUANT SUPERDAY / TOPIC B

An alpha factory built to reject its own output

An LLM agent loop that invents economic theses, implements them as formulas, and then spends most of its effort trying to prove them wrong.

Universe 200 most liquid Indian equities

Span 2014-01-01 to 2025-12-31

Data 4,988,593 price rows

Tests 315 passing

One finished unit of work is an Alpha Card

An alpha signal is one number per stock per day. Only the ranking matters, so RankIC is the grade.

IN feature panel, point-in-time | universe | deterministic backtester | LLM token budget B | research corpus | production factor book

OUT one Alpha Card

THE EIGHT PARTS OF A CARD

- 1 **Economic thesis** - mechanism, counterparty, why not arbitrated
- 2 **Pre-registered sign** and horizon, hashed and timestamped
- 3 **Formula** over causal operators, with its canonical AST
- 4 **Backtest report** - RankIC, ICIR, Sharpe, turnover, MDD, decay
- 5 **Honesty audit** - marginal IC, Deflated Sharpe, t, PBO, peek spent
- 6 **Red-Team report** - which attacks ran, and each result
- 7 **Verdict and lineage** - thesis, parent formula, edit, generation
- 8 **Provenance** - fields used, read off the AST leaves

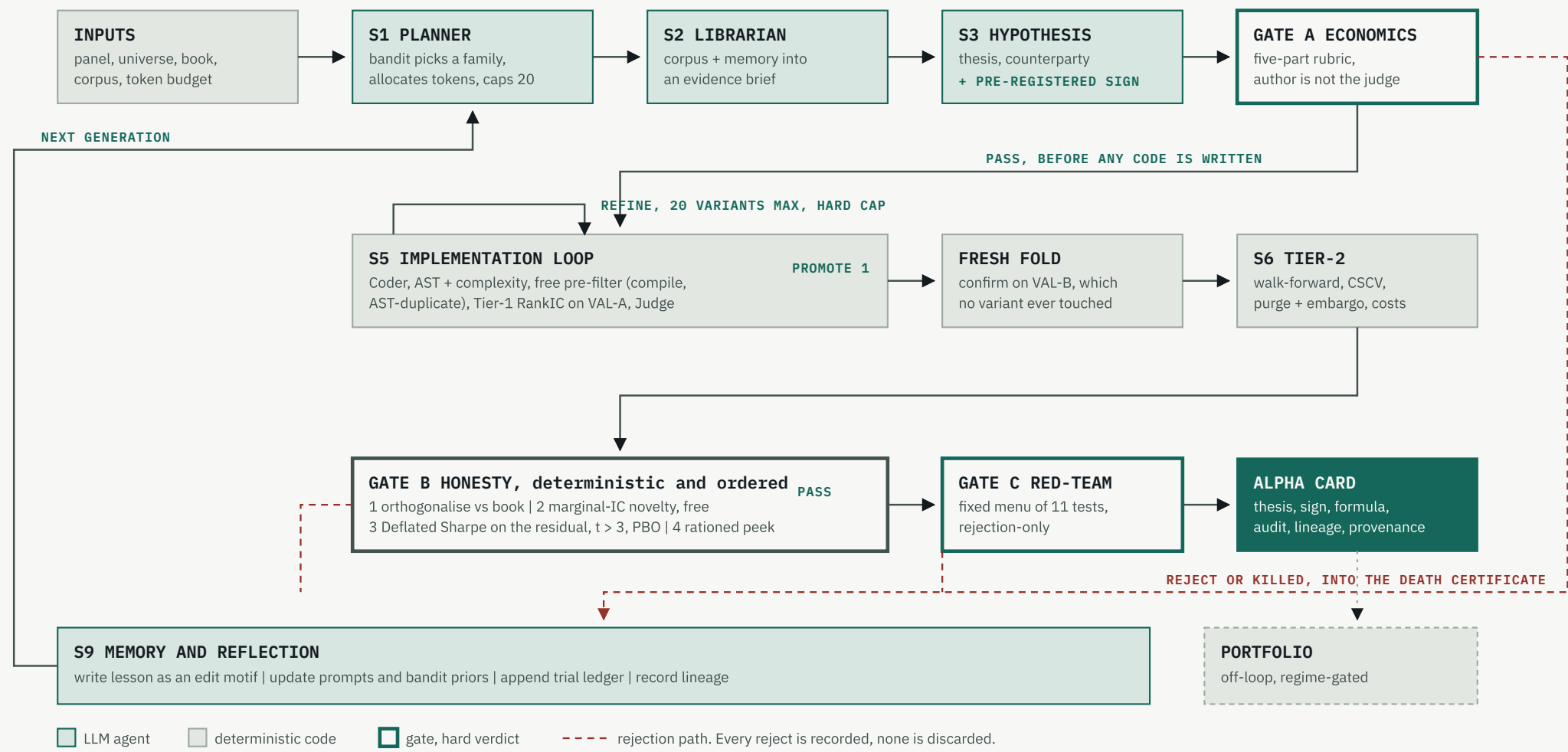
MEASUREMENT CONTRACT

Label	next-day return, cross-sectionally demeaned
Timing	features to close of t, trade t+1 open, earn t+1 open to t+2 open
Primary	RankIC, with IC and ICIR alongside
Decay	h in {1, 2, 3, 5, 10, 21}
Costs	15 bps per side, swept 5 / 15 / 30 - robustness, not the grade

A rejected card is not discarded. It carries its death certificate into memory, and that is the substrate for improvement across generations.

Nine stages, four gates, one feedback loop

Agency where there is a **decision**. Deterministic code where it is a **fixed computation**. Every verdict threshold is code.



Gate ordering is deliberate. **Gate A** kills before any code is written. The **pre-filter** inside S5 is free, so it costs zero trials. **Gate B** checks novelty before statistics, so a scarce holdout peek is never spent on a clone. **Gate C** runs last because it can only reject, never promote.

A universe defined by a rule, not by a list

Everything downstream is arithmetic on this panel. If it is biased, every later result is quietly too good.

IN

NSE daily bhavcopy, 2014 to 2025 | NSE corporate-action feed | NSE industry list

OUT

monthly 200-name universe | adjusted price panel | feature panel with demeaned labels

The rule is the universe. Each month, take the 200 names with the highest trailing 63-day median turnover, requiring 252 days of history, from that month's own bhavcopy. The rule reads only the past, so a stock is in it exactly while it was liquid, **including the ones that later went bankrupt**.

2.6e-17 Coverage slope, members per day. Yearly mean is 200.0 every year, 2015 to 2025. No survivorship slope.	72 / 72 Month-ends bit-identical after truncating the data at 2020-01-01, and again at 2022-06-30. No look-ahead.	100.0% Of 2,697 days on which RELIANCE, TCS, SBIN, TATASTEEL, MARUTI and ONGC are all present.	2019-12 DHFL leaves the universe when its turnover collapsed, not when it delisted. The correct behaviour.
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WHAT WAS BUILT UNDERNEATH IT

Prices	4,988,593 rows, 3,178 symbols, 2,961 trading days. Keyed by ISIN, not ticker , so a rename stays one continuous series
Corporate actions	26,325 events parsed. 830 splits and bonuses back-adjusted across 596 symbols, anchored to the observed price break rather than trusting the announced ex-date
Cross-check	daily-return correlation against yfinance on the 30 most liquid names: median 0.9963 , 25 of 30 above 0.99. The five below are explained name by name
Panel	539,400 rows. Ten features plus a size proxy and sector. Labels demeaned cross-sectionally every day
Leak self-test	shift the whole panel forward one day and a known factor's RankIC must move. Shift it backward and the leak signature appears: -0.3405

DOES THE PANEL CONTAIN SIGNAL? MEAN DAILY RANKIC ON VAL-A, IR IN BRACKETS

FEATURE	H=21	READS AS
vol_21	-0.082 [-11.9]	low-volatility anomaly
max_ret_21	-0.072 [-14.6]	lottery, or MAX, effect
mom_126	+0.063 [+9.1]	6 minus 1 month momentum
mom_21	-0.013	1-month momentum is genuinely near zero in India

DISCLOSED DEFECTS, ALL OF THEM

- **delivery_pct starts 2019-10-01.** Zero coverage in TRAIN. Left as NaN, never fabricated
- **Dividends and demergers are not price-adjusted.** Nine jumps flagged and kept verbatim, not winsorised
- **No point-in-time fundamentals.** Free Indian fundamental data is not PIT, so using it would inject the exact look-ahead this system exists to catch. A rigor choice, not an oversight

Planner and Librarian: decide where to look next

The only stage that spends token budget deciding **what not to try**.

IN

bandit state over 10 idea-families | past lessons | remaining token budget | 53-entry research corpus

OUT

chosen family, token allocation, **variant budget**, evidence brief with citations

WHAT EACH ONE DECIDES

- **S1** allocates tokens by softmax over realised reward per family
 - **5 percent exploration floor**: no family can ever be starved to zero
 - **S1** writes the variant budget that S5 is forced to obey
 - **S2** retrieves by family plus keyword. No embeddings, **no vector DB**
 - Non-tradeable anomalies are removed **by code, not by the LLM**
 - Stop rule: token exhaustion, then diminishing returns, then a hard cap
- Why no vector database.** A few hundred lessons. Numpy cosine is instant, and a vector DB is one more dependency that can break during a live demo.

0.0574

Share still held by the worst family after 50 consecutive failures

17 / 53

Corpus anomalies flagged **not tradeable** with our data, and excluded

THE CORPUS

53 anomalies	across 10 idea-families
Sources	paper abstracts and published anomaly lists
Each entry	mechanism, counterparty, horizon, tradeable flag
Retrieval	exact family match, then keyword hit count

Measured. Ask the Librarian for a fundamental-data family and it returns **zero suggested angles** with every candidate listed under "excluded from suggestions". The agent is not trusted to remember what it cannot trade.

Hypothesis and Gate A: commit to a story, and to a direction

The cheapest possible rejection, and the only place a prediction can still be made honestly.

IN evidence brief | idea family | prior lessons for this family

OUT thesis + pre-registered sign, sha256-hashed into the ledger, then pass or reject

GATE A RUBRIC: ALL FIVE, OR REJECT

- 1 A named **mechanism**
- 2 Who the **counterparty** is, and why they persistently lose
- 3 Why it has **not already been arbitrated** away
- 4 **Horizon** and **regime**
- 5 A **falsifiable** prediction, including the sign

THREE TEETH

- The rubric is hard, not advisory
- A **different LLM client** scores it. The author never grades itself
- The sign is hashed now and checked against the realised sign later

INNOVATION: THE PRE-REGISTERED SIGN

Before any data is touched, the thesis is serialised to canonical JSON, **sha256-hashed and timestamped** into the trial ledger.

A signal that works only with the **opposite** sign to its own story is a **thesis failure, not a discovery**, and the mismatch never flips the sign to rescue it.

AlphaAgent aligns hypothesis and factor *post hoc*. AgonAlpha's reviewer audits sign logic *after* evaluation. We found nothing that commits a direction *before* evaluation and rejects on mismatch. This is our lead claim.

hash < bt

Prereg hash is written at a strictly earlier event index than the first backtest. Asserted.

0 calls

A blank rubric field is rejected structurally, with no LLM call spent at all

Author is not the judge, asserted by object identity. The test checks `economics.client` is not `hypothesis.client`, so a refactor cannot quietly collapse them into one conversation.

Implementation: bounded formula search

The thesis says "volume shock". It does not say whether the window is 5, 10 or 20 days. Only data can answer that, and that is exactly why it is dangerous.

INthesis + sign | 35 causal operators | 35-formula zoo for duplicate detection | feature panel

OUTone promoted formula, canonical AST, complexity, Tier-1 metrics, and up to 20 ledger rows

THE INNER LOOP

Coder > AST and complexity > **free pre-filter** (compile check, AST-duplicate vs zoo and memory) > Tier-1 RankIC on VAL-A > Judge > refine or promote.

THREE BINDINGS

- 1 **Hard cap of 20 variants per thesis**, enforced by the Planner
- 2 **Every variant enters the trial ledger**, and near-identical ASTs are clustered so 20 tweaks are not 20 independent bets
- 3 **Fresh-fold confirmation**. Search runs on Train + VAL-A. The winner is confirmed on VAL-B, which no variant ever touched, so this costs no holdout peek

This turns "we do formula search" from a liability into a rigor point: **bounded, disclosed, deflated, and confirmed out of sample**.

20

Exact measured maximum with a Judge injected to say "refine" forever

0

Trials charged by the pre-filter. Junk dies for free, before the Judge is paid

THE OPERATOR LIBRARY IS THE SAFETY PROPERTY

35 operators	13 strictly trailing time-series, 5 cross-sectional, 17 element-wise
13 of 13	time-series operators covered by a causality test: perturb a future row, assert every earlier output is bit-identical
Parser rejects	__import__, attribute access, subscripts, comprehensions, lambdas, f-strings
Zoo	25 Alpha101 + 10 classical, canonicalised and fingerprinted for duplicate detection

No operator can reach forward. Formula-level look-ahead is not policed here, it is **structurally impossible**. That is also why arbitrary Python factor evolution is excluded, not merely deferred.

Backtester: one engine, eight call sites, no decisions

The measuring instrument. It computes and returns a fixed dict. It never promotes, rejects or interprets.

IN

signal (long or wide) | split | horizon | extra lag | cost bps | neutralize | subsample

OUT

Metrics: rank_ic, ic, icir, t_stat, sharpe, ann_return, turnover, mdd, n_days, n_obs, decay, sign

WHAT IT GUARANTEES

- **Eight call sites, one engine:** Tier-1, Tier-2, holdout, stress, cost sweep, curriculum, portfolio, ablation
- **Purge and embargo** around every train and test boundary
- **HOLDOUT raises PermissionError** unless a ledger peek token is presented
- A HOLDOUT-tail purge stops a VAL-B label reaching across the seal
- **Regime labels use expanding-window thresholds only.** A 2016 day cannot know 2024's volatility
- No RNG. Two identical calls return bit-identical dicts, nested decay curve included

A look-ahead we found in our own engine. Regime labels originally used full-sample quantiles. Fixed at source, not patched in the test. A companion test confirms the broken version **would** have been detected, so the check is not vacuous.

DOES THE INSTRUMENT WORK? FIVE CALIBRATION CHECKS

CHECK	EXPECTED	MEASURED
Random noise gives no signal	IC < 0.01	0.0023
Known planted signal recovered	0.0400	0.0404
A label predicting itself	> 0.9	1.0000
...then given one extra day of lag	collapse	-0.0052
Negating a signal negates its IC	exact	bit-exact

2.3 s	-38.7
Full backtest on the real panel, 581 symbols, 857 evaluation days	Sharpe at 80 bps, from +4.14 at zero. Strictly decreasing in cost

Why this matters for the whole design. Every gate downstream is arithmetic on this dict. If the instrument is wrong, the deflation, the PBO and the red team are all rigorous statements about a fiction.

Gate B: is it new, and is it real?

Deterministic thresholds only. The **order** of the four steps is load-bearing, not cosmetic.

IN promoted formula | production factor book | the whole trial ledger | pre-registered sign

OUT **verdict + honesty audit:** marginal IC, DSR, t, PBO, effective N, peek id, what the peek scored

FOUR ORDERED STEPS

- 1 Orthogonalise** against the book, per day, cross-sectionally. Keep the residual
- 2 Novelty:** marginal IC of the residual. **Free**, so a clone dies here having spent nothing
- 3 Statistics:** Deflated Sharpe on the residual, $t > 3$, PBO, deflated by the **run-wide effective** trial count
- 4 Rationed holdout peek**, also on the residual, so what was spent is auditable

MEASURED ON OUR FIXTURES

Best of 200 pure-noise signals	DSR 0.477	REJECT
Planted signal found in 5 trials, $t = 7.065$	DSR 0.9952	ACCEPT
20 formulas differing only by window length	N 20 to 2.0	clustered
PBO, 300 noise matrices	0.4859	is 0.5
PBO, one real signal among eleven	0.0019	is 0
Holdout peeks, then None forever	12	append-only
The peek scored the residual, not the raw	0.01957	raw 0.03200

FOUND IN OUR OWN BUILD

We searched **40 pure-noise variants** across 40 different thesis IDs, then gated the winner under a fresh 41st thesis. Its raw t-statistic came out at

$t = -3.000$

Pure noise, clearing the celebrated $t > 3$ bar exactly.

Deflation scoped to the thesis	N=1	DSR 1.000	ACCEPT
Deflation scoped run-wide	N=41	DSR 0.789	REJECT

The deflator is only as honest as the counter behind it, and every number the broken gate printed looked impeccable.

The ledger. Append-only SQLite. There is no DELETE, DROP TABLE or TRUNCATE anywhere in the module, and a structural scan of the source asserts it.

Gate C: try to kill it

The agent chooses the attacks. Code runs them. Every test is a parameterised backtest, never LLM-written code.

IN

surviving formula | claimed horizon and regime | expanding-window regime labels | monthly liquidity ranks

OUT

survives or killed, per-test result, flagged diagnostics, and zero added trials

THE FIXED MENU OF ELEVEN. FIVE ARE DECISIVE AND ALWAYS RUN.			
#	TEST	KILL RULE	KIND
1	subsample_year	drop the best year; RankIC collapses and loses significance	DECISIVE
2	regime_split	RankIC at or below zero in bull or in bear	DECISIVE
4	cost_sweep	net loss at 15 bps, or Sharpe halved to under 0.5	DECISIVE
5	extra_lag	one more trading day of lag destroys it	DECISIVE
10	sign_stability	modal sign holds in under 70 percent of folds	DECISIVE
3	size_tercile	by trailing turnover, not market cap , which would leak	diagnostic
6	delivery_lag	shift only the one field with real timing ambiguity	diagnostic
7-9	sector_neutral, liquidity_filter, decay_curve	is it a sector bet, does it survive above the 40th turnover percentile, does the edge live at the claimed horizon	diagnostic
11	universe_edge	drop the names ranked 150 to 200 by liquidity that month	diagnostic

INNOVATION: REJECTION-ONLY, AND WE COUNT IT

A filter that can kill but never promote **cannot inflate the false-discovery rate**, so it needs no deflation. That is the answer to "does your red team running eleven backtests blow up your trial count?"

Measured: a full run made **25 backtests**, wrote **25 ledger rows**, every one with `counts_as_trial = 0`. Afterwards `ledger.n_trials() == 0`.

WHAT IT ACTUALLY KILLS		
PLANTED PATHOLOGY	BEFORE	AFTER
Leakage: a signal that is its own label	1.0000	0.0018
One lucky year, noise elsewhere	-	0.0019
High turnover, thin gross edge (Sharpe)	0.415	-7.97
A sign that flips between years	-	0.60
A clean AR(1) signal (RankIC)	0.0357	11 of 11

The last row matters as much as the others. A red team that kills everything is not a filter, it is a wall.

Memory and Reflection: learn from the corpse

Rejected cards are the majority of the output, so they are the majority of the learning signal.

IN

the card, **accepted or rejected** | verdict and reason | metrics | red-team result

OUT

lesson as an edit motif, bandit prior update, lineage row, ledger row, revised prompts

SIX STORES		GUARDS, SO MEMORY CANNOT BECOME SUPERSTITION	
Formula index	exact hash plus fingerprint bucket	Confidence gate	A motif seen once is not a prior . At three observations it is.
Lesson store	edit motifs. Physically a separate file from the exact stores	Asymmetric veto	Dead in momentum, still live in reversal. Context-scoped and sticky .
Bandit state	10 families, pulls and rewards	Exploration floor	Five percent, so a bad run cannot permanently close a family.
Card store	human-readable JSON, plus a queryable index	Curriculum	Rotates a mandatory adversarial regime every N generations.
Lineage tree	parent pointers, walkable root to leaf	Meta-check	If rolling holdout FDR rises, gate thresholds auto-tighten .
Accepted book	the panel Gate B orthogonalises against		

INNOVATION: MOTIFS, NOT OUTPUTS

Memory stores **which change helped, under which parent context** - "shortening the window helped, in momentum" - rather than storing finished formulas. A finished formula teaches you one answer. A motif teaches you a move.

1 -> 0

Priors returned at one observation. At three, one prior is returned.

3.0 up

t-stat bar when FDR hits 0.9. At FDR 0.0 it stays exactly 3.0.

Budget safety. The LLM call happens first, so exhausting the token budget raises **before** anything is written. Measured: memory and bandit are left completely untouched.

Five failure modes, five mechanisms, and what each does not catch

The design's real content is knowing which mechanism catches which failure, **and which one it sails straight past.**

FAILURE MODE	CAUGHT BY	DOES NOT CATCH IT
Look-ahead and leakage	Causal operator library (structural), timing contract, purge and embargo, extra-lag test, delivery shift	Deflated Sharpe and PBO. A deliberately leaky oracle at Sharpe 35 survives both. Gençay, arXiv 2608.27734
Selection and multiple testing	Deflated Sharpe vs the effective trial count, PBO, variant cap, fresh fold, rationed holdout	Any amount of economic reasoning. A beautiful story does not reduce your trial count
Story-fitting right answer, wrong reason	Pre-registered sign , counterparty rubric, author is not the judge	IC, Sharpe and DSR all pass happily. A signal with the wrong sign for its own story is statistically excellent
Redundancy and crowding	AST duplicate check in S5, then marginal IC against the production book in Gate B	All of the above. A perfect clone of an incumbent factor passes every statistical and economic test there is
Regime and cost fragility	Red-Team fixed menu: regime split, cost sweep, size tercile, liquidity filter, universe edge	Everything measured in-sample. A full-period RankIC averages a bull and a bear into a number describing neither
Broken data source	External reconciliation only. We found one. Slide 13.	Every mechanism in this table. It contaminates the universe, not any one factor. Nothing throws an exception

Formula-level look-ahead is structurally impossible. Every operator is trailing-window or same-day cross-sectional. No operator can reach forward, and all 13 time-series operators carry a causality test that proves it.

Statistical gates catch over-searching. They do not catch cheating. Different failures, different machinery. Deflating a leaky signal by its trial count gives you a rigorous number about a fiction.

Three budgets, and the conflict between the first and the third

The prompt names the token budget. It is real, but it is not the scarce resource that decides whether a discovery is true.

BUDGET	SPENT BY	IS A BACKTEST EXPENSIVE HERE?	HOW WE ECONOMISE
1. LLM tokens	Hypothesis, Coder, Judge, Red-Team, Reflection	No. A backtest is a Python call returning a small dict. Near zero tokens	Bandit funds productive families. The free pre-filter kills junk before we pay a Judge
2. Compute	Running backtests	Barely. The Tier-2 CSCV battery is modestly heavier	Heavy CSCV only on finalists
3. Statistical integrity	Every selecting backtest is one trial. Every holdout peek is irreplaceable	Yes, invisibly. Nothing gets slower, nothing errors. The bar for everything else just rises	Variant cap, append-only ledger, Deflated Sharpe, 12 peeks ever

THE CONFLICT. WE HAVE NOT FOUND THIS STATED ANYWHERE.

Budgets **1 and 3 pull in opposite directions**. Token efficiency rewards finding the best candidate in **fewer** evaluations. Statistical integrity punishes you for **every** evaluation made along the way.

Any technique that makes search more sample-efficient at maximising a noisy objective is, by the same token, more efficient at overfitting.

This is why MCTS is on our roadmap rather than in our loop.

EVIDENCE FOR BUDGET 3: SEARCH N WORTHLESS CANDIDATES, TAKE THE BEST

VARIANTS N	SQRT(2 LN N)	REALISED BEST T	P(BEST T > 3)
20 our cap	2.45	1.87	2.6%
200	3.26	2.75	23.7%
500	3.53	3.04	49.2%

200,000 Monte-Carlo searches per row, seed 42. At 500 variants a pure-noise search clears the t > 3 bar **half the time**, so the bar carries no information at all.

The flip side is our best argument. Theory-first, LLM-guided search tries fewer and smarter candidates than brute-force mining. Fewer trials, smaller deflation penalty, more believable survivors. **The economic thesis is what buys the statistics their power.**

Three ways to be wrong: data, statistics, economics

Told in three beats each. The first is not a demonstration. It happened to us, on the data we were handed.

1. DATA INTEGRITY. REAL.

The universe source was structurally broken

NAIVE	The supplied constituent file passes every superficial check: 37 snapshots, exactly 200 names in each , dead companies retained, inclusion and exclusion events tracked.
CAUGHT BY	External reconciliation against NSE, not by any gate. 42 stocks that were already trading in 2015 and sit in today's most-liquid 200 never appear in any snapshot : RELIANCE, TCS, SBIN, ONGC, MARUTI, TATASTEEL, COALINDIA, POWERGRID, SUNPHARMA, TITAN. 100 of 315 names carry zero events. The signature of a change-log replayed onto an incomplete seed, then padded back to 200 with mid-caps.
WHY IT LANDS	The missing names are systematically the largest and most liquid in India . Every liquidity, size and capacity feature would have been computed on a sample with its top end sawn off. DSR, PBO, purge and the lag test all pass it silently.
FIX	Abandon constituent data entirely. Rebuild from daily bhavcopy as the top 200 by trailing turnover: point-in-time and survivorship-free by construction. Coverage slope 2.6e-17 per day, and it recovers all six heavyweights on 100 percent of days.

2. STATISTICAL

Look-ahead leakage

NAIVE	A factor that lets same-day or forward information into the return window posts a spectacular Tier-1 RankIC. A signal that is quietly its own label scores 1.0000 .
CAUGHT BY	Structure, three times over. The causal operator library makes it inexpressible. The panel self-test detects it and shows it collapsing to -0.060 one day later. Red-team test 5 destroys it: 1.0000 to 0.0018 on one extra day of lag.
TEACHING POINT	The Deflated Sharpe would have passed it. Gençay's leaky oracle posts Sharpe 35 and survives DSR and PBO completely. Statistical gates catch over-searching. They do not catch cheating.
FIX	Close the surface rather than measure it. Causal operators only, an explicit timing contract, purge and embargo, and a lag test that can kill on its own. It is also why arbitrary-Python factor evolution stays out of scope.

3. ECONOMIC

Right answer, wrong reason

NAIVE	A data-mined signal posts a respectable IC and passes every statistical gate, but its realised sign is the opposite of what its own thesis predicted. The story said crowded stocks underperform. The money was on the other side.
CAUGHT BY	The pre-registered sign. The thesis is hashed and timestamped before any data is touched, and a test asserts the hash lands at a strictly earlier event index than the first backtest.
VERDICT	Rejected as a thesis failure, not a discovery , and the sign is never flipped to rescue it . That is the tempting move, and refusing it is the whole point. Costs zero holdout peeks .
TEACHING POINT	No purely statistical gate would ever flag this. IC, Sharpe, DSR, PBO and the red team all pass it. It is the one failure mode where the economics is the test.

Status. Example 1 is a real finding on real supplied data, reproducible from the CSV in the repo. Examples 2 and 3 are demonstrated at component level with the measured values shown. The full loop run that emits them as finished rejected Alpha Cards has **not been run yet**. See slide 16.

Grading the factory, not the signal

Grading one signal is the Alpha Card. Grading the factory asks entirely different questions.

FIVE DIMENSIONS

Yield	hypotheses per accepted card, tokens per accepted alpha , marginal IC added per generation, diversity
Honesty	FDR = accepted but fails holdout, over accepted. DSR distribution. Sign agreement rate
Efficiency	real alpha per token. The headline objective
Gate value	the ablation: catch rate, false-kill rate, FDR with the gate on and off
Real learning	error volume trend, not error type

DETECTING FAKE LEARNING

Factor-mining agents can look like they are learning while only their **mistakes** mature: conceptual, then operational, then strategic, with total error volume barely moving.

So we track **rejections per generation** and per-gate pass rate, not the changing character of failures. Falling volume is improvement. Same volume in new clothes is drift. **We will report which one we see, including if it is the unflattering answer.**

THE ABLATION: DESIGNED AND SPECIFIED NOT YET RUN

Seed a pool of about 40 factors with **documented ground truth**: 10 genuinely predictive, 10 pure noise, 10 deliberately overfit, 10 deliberately leaky. Run the pool with each gate **enabled, then disabled**.

Report per gate: **catch rate, false-kill rate**, and headline **FDR on versus off**.

This makes complexity **self-justifying rather than asserted**. We did not add gates because papers have them. We measured what each one catches. The corollary is the real argument for pruning: **we can only make that argument for gates we actually ablated.**

WHAT WE CANNOT YET SHOW, AND WILL NOT PRETEND TO

- **No accepted Alpha Card exists.** The full accept path is exercised end to end on a fixture, but not on a real thesis
- **No ablation numbers.** Designed above, none measured. There are no placeholder figures anywhere in this deck
- **No fake-learning plot.** It needs multi-generation run data that does not exist yet
- Small samples will make these numbers **illustrative, not conclusive**, and we will label them that way

Honest novelty: two claims conceded, two that survive

Two of our four original headline claims were already published. Conceding them with citations is a stronger position than a bluff.

CLAIM	STATUS	POSITION
Pre-registered sign and counterparty gate	GENUINELY NOVEL	Our lead claim. AlphaAgent does post-hoc semantic alignment. AgonAlpha's reviewer audits sign logic after evaluation. Nothing found commits a direction before evaluation and rejects on mismatch . A post-hoc check can be talked around. A hashed commitment cannot.
Three budgets, and their conflict	NOVEL	Token efficiency and statistical integrity pull in opposite directions. Not found stated anywhere, and it produces a falsifiable consequence: MCTS goes on the roadmap, not in the loop.
Fixed-menu, rejection-only red team	DIFFERENTIATED	Adversarial review already exists in AgonAlpha and FactorMAD. Ours is a fixed, parameterised menu , so no arbitrary code runs and every attack reproduces, and it is provably rejection-only , so it cannot inflate the trial count.
Statistical-rigor gates wired into fitness	ANTICIPATED	Not novel. Conceded. Gençay, arXiv 2608.27734 , got there first. We cite it and lead instead with what DSR does not catch : their leaky oracle posts Sharpe 35 and survives DSR and PBO completely. Statistical gates are table stakes now. Knowing their blind spot is the contribution.

LIMITATIONS, STATED BEFORE YOU FIND THEM

- **No accepted card and no ablation numbers yet.** The two deliverables that need a full loop run
- **One market, one universe, prototype scale.** No evidence about production breadth
- **No point-in-time fundamentals**, so whole feature families are unavailable. A rigor choice that costs coverage
- **CSCV applies a light purge, not a full embargo.** Strategies are not refit inside splits, so leakage is limited to label overlap at block seams
- **The LLM tier is a free-tier model chain**, not a frontier model. The architecture is model-agnostic. Thesis quality is not

ROADMAP, AND THE PRECONDITION FOR EACH

EXTENSION	PRECONDITION
LLM-guided MCTS formula search	Trial accounting validated first. MCTS's adaptive draws make effective independent trials hard to define
Code-based factor evolution	A causal-operator sandbox. Arbitrary Python reopens the look-ahead surface we closed
Point-in-time fundamentals	Vendor access. This is where a per-field availability registry finally earns its keep
Full evolutionary run, live-forward test	Time

Get the meter working before attaching the multiplier.

What exists, and a dashboard to interrogate it

Every phase was human-verified before the next began, against a report requiring a **measured value for every criterion**.

315 tests passing, plain pytest, no network, 12m53s	11,147 lines across 16 source modules	10 / 13 system phases built and signed off	14 dashboard pages, one per subsystem
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MODULE	WHAT IT IS	STATE
universe, prices, panel	Liquidity-defined universe, ISIN-keyed NSE price build with our own corporate-action adjustment, 539k-row feature panel	VERIFIED
backtester	One engine, eight call sites, purge and embargo, holdout lock, expanding regime labels, bit-identical repeats	VERIFIED
operators, ast_tools, zoo	35 causal operators, whitelist parser, canonicalisation and fingerprinting, 35-formula zoo	VERIFIED
gates, ledger	Deflated Sharpe, effective trial count, PBO via CSCV, walk-forward, ordered Gate B, append-only ledger with a 12-peek ration	VERIFIED
memory	Six stores. Exact and semantic stores are physically separate files	VERIFIED
agents	Eight agents on one call path: model probe with fallback chains, throttle, budget, prefix caching, schema repair, offline mock mode	VERIFIED
redteam	11 falsification tests, 5 decisive, all recorded as non-trials	VERIFIED
loop	17-node orchestration graph, stdlib-only SQLite checkpointer, resume reproduces an identical state digest	VERIFIED
demo run, evaluation	The accepted card, the bad examples as finished cards, and the gate ablation	PENDING

THE DASHBOARD

streamlit run dashboard/Home.py

Two jobs. Explain the machine, with flowcharts and prose. Then let a reviewer poke at the evidence: the survivorship-free universe, the coverage plot with its fitted slope, the panel, the trial ledger, the red team, the cards.

Where it matters it computes live rather than showing a picture. Run a backtest, run the red team, compute a Deflated Sharpe against a trial count you choose.

Read-only over the project's data. No LLM call, and **no network on any page load**.

Why the verification protocol is on a slide. This system exists to catch self-deception in signals, so the same discipline has to apply to the build. **A survivorship-biased panel does not throw an exception. It just makes every later result too good.** The t = -3.000 noise winner and the full-sample regime labels both surfaced this way, in code whose own tests were already green.

What we took from each paper

Every entry re-verified against arXiv or the ACM DL, September 2026. **Bold** marks what this system actually adopted or rejected.

AGENTIC ALPHA MINING

AlphaAgent: LLM-Driven Alpha Mining with Regularized Exploration to Counteract Alpha Decay. Tang, Chen, Yang, Mai, Zheng, Wang, Chen, Lin. [arXiv 2502.16789](#)
Adopted: the idea to factor to evaluation spine, **AST-based originality**, AST complexity control. **Differ:** their hypothesis-factor alignment is post-hoc; ours is a sign committed before evaluation.

R&D-Agent-Quant: A Multi-Agent Framework for Data-Centric Factors and Model Joint Optimization. Li, Yang, Yang, Xu, Wang, Liu, Bian. *NeurIPS 2025*. [arXiv 2505.15155](#)
Adopted: the research, development, feedback loop, and the **multi-armed-bandit scheduler** that allocates budget across idea-families. That is our S1.

AgonAlpha: Autonomous Alpha Discovery via Prompt Economy and Scalable Agentic Search. Ye, Sun, Ren, Yu, Yi, Yang. [arXiv 2608.11250](#)
Conceded to: the fresh-context adversarial reviewer with veto. Our Gate A separation is the same pattern. **Contrast:** it uses no DSR, PBO or purge and embargo, which is the gap Gate B fills.

AlphaMemo: Structured Search-Process Memory for Self-Evolving Alpha Mining Agents. Yu, Zheng, Pan, Liu, Wang, He. [arXiv 2606.20625](#)
Adopted: memory as **AST-diff edit motifs**, which change helped in which parent context, rather than final outputs. Plus confidence-gated retrieval and the asymmetric veto.

FactorMAD: A Multi-Agent Debate Framework Based on LLMs for Interpretable Stock Alpha Factor Mining. Duan, Zhang, Li. *ICAIF 2025*. doi [10.1145/3768292.3770377](#)
Adopted: adversarial multi-agent critique as a quality mechanism. **Rejected:** its code-based factor pipeline. Arbitrary Python reopens the look-ahead surface.

QuantAgent: Seeking Holy Grail in Trading by Self-Improving LLM. [arXiv 2402.03755](#) / **AlphaGen: Synergistic Formulaic Alpha Collections via RL.** [arXiv 2306.12964](#)
Adopted: the inner writer-judge loop inside an outer knowledge-base loop, which is our S5 plus S9; and the synergistic-collection framing behind the off-loop portfolio step.

STATISTICAL RIGOR AND HONEST EVALUATION

What survives honest evaluation? Leakage-safe, search-aware assessment of LLM-driven trading strategy discovery. Eray Gençay. [arXiv 2608.27734](#)
Conceded to on wiring statistical rigor into fitness. **Adopted:** its central result, **a deliberately leaky oracle at Sharpe 35 survives DSR and PBO**. That is the top row of our failure-mode matrix.

The Deflated Sharpe Ratio: Correcting for Selection Bias, Backtest Overfitting and Non-Normality. Bailey, Lopez de Prado. *SSRN 2460551*
Adopted: the expected-maximum-Sharpe deflator, skew and non-excess kurtosis, and the insistence that the trial count is part of the result. Ours is checked against a Monte-Carlo order statistic to 0.033.

Harvey, Liu and Zhu. "...and the Cross-Section of Expected Returns". *Review of Financial Studies, 2016*
Adopted: the **t > 3** hurdle, and the factor-zoo framing that makes marginal IC against an incumbent book the right novelty test.

Lopez de Prado. Advances in Financial Machine Learning
Adopted: CSCV into PBO, purging and embargo, and the discipline that backtesting is a final confirmation rather than a research tool. That is why VAL-B is touched once.

The Alpha Factory Illusion. *LLMQuant*
Adopted: the fake-learning test. Track error **volume** per generation, not the maturing character of errors, because sophistication is not improvement.

ValueBlindBench: Agreement-Gated Stress Testing of LLM-Judged Investment Rationales. [arXiv 2604.25224](#)
Adopted: the requirement to validate that an LLM judge is not rubber-stamping. Hence the author-is-not-judge object-identity assertion and Gate A's structural pre-check.

OPERATORS, SEARCH AND DATA

101 Formulaic Alphas. Zura Kakushadze. [arXiv 1601.00991](#). *Wilmott 2016(84):72-80*
Adopted: the operator vocabulary and 25 of the 35 zoo formulas. **Disclosed:** Alpha 56 is skipped, since it needs true market capitalisation and our size proxy is trailing turnover.

Alpha Jungle: An LLM-Powered MCTS Framework for Formulaic Factor Mining. Shi, Duan, Li. [arXiv 2505.11122](#) / **CogAlpha: Cognitive Alpha Mining via LLM-Driven Code-Based Evolution.** Liu et al. [arXiv 2511.18850](#)
On the roadmap, deliberately not in the loop. Both make search more sample-efficient at maximising a noisy objective, which is the budget 1 versus budget 3 conflict on slide 12.

Data: NSE daily bhavcopy, NSE corporate-action feed, NSE total-market industry list. yfinance used only as an independent cross-check
No paid data source and no paid API anywhere in the build. Raw downloads are cached verbatim and never modified in place.